

DELHI PUBLIC SCHOOL, BOKARO STEEL CITY

Class X — Biology — Unit Test

Time: 1 hour

Maximum Marks: 47

Answer all questions. Marks are shown in brackets.

1. Which blood vessel carries blood away from the heart? [2]
2. Which of the following organelles is primarily involved in photosynthesis? [2]
3. Explain the role of chloroplasts in photosynthesis, naming the main pigments involved and briefly outlining the two major stages of the process. [2]
4. Describe the flow of blood through the human heart, starting from the right atrium and ending at the aorta. Include the names of the valves crossed. [2]
5. Draw a labelled diagram of an alveolus showing capillaries, air space and the direction of gas exchange. [2]
6. Draw a neat labelled diagram of the human digestive system (stomach, small intestine, large intestine, liver, pancreas) and label the site where most absorption occurs. [5]
7. Draw and label a nephron (Bowman's capsule, glomerulus, proximal tubule, loop of Henle, distal tubule, collecting duct). [5]
8. Explain the structural differences between palisade mesophyll and spongy mesophyll, and state how each structure aids its function in the leaf. [5]

9. Describe the process of transpiration in two to three sentences and name two environmental factors that increase its rate. [5]
10. Explain how the structure of xylem vessels facilitates water transport in plants, mentioning one structural feature and what it does. [5]
11. (a) A diagram shows two potted plants — Plant A is in bright light with broad green leaves, Plant B is kept in dim light with pale, elongated leaves. State one reason for the difference. [2]
11. (b) Suggest one practical measure to help Plant B recover. [3]
12. A resting person has a tidal volume of 0.5 L and breathes 12 times per minute. Calculate the pulmonary ventilation rate. [4]
13. If dead space is 0.15 L per breath, calculate the alveolar ventilation per minute. Show your working. [5]